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SLOT - D

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SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%

Code of Conduct:

I S. Nirmal (192511249) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S. Nirmal

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks	Q5
Understanding of Scenario	20%	2	1	1	1	2
Identification of Key Issues	20%	1	1	1	1	1
Application of Concepts	25%	1	1	1	1	1
Decision Making	20%	1	1	1	1	1
Clarity of Response	15%	1	1	1	1	1

27/30

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → **/24**

Quality Assurance → **/25**

Human Resources → **/26**

Executive Management → **/27**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

3.A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the **/64** prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.

2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.
4. The **default route** that should be used if the destination does not match any routing table entry.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Scenario	20%	Demonstrates clear and complete understandin	Shows good understandi ng with minor gaps	Partial understandin g; misses key aspects	Misinterpret s or fails to understand the scenario

Answer 1: Smart Campus Subnet Design for 192.168.100.0/24

1. Understanding of the Scenario

A new Smart Campus must host three academic departments on one private address block, **192.168.100.0/24** (256 addresses). The masks are already fixed: **School of Computing → /25**, **School of Electronics → /26** and **School of Mechanical Engineering → /27**. As the network engineer, I must work out the exact address boundaries of each department, so that the subnets do not overlap, and report the address space that is still free.

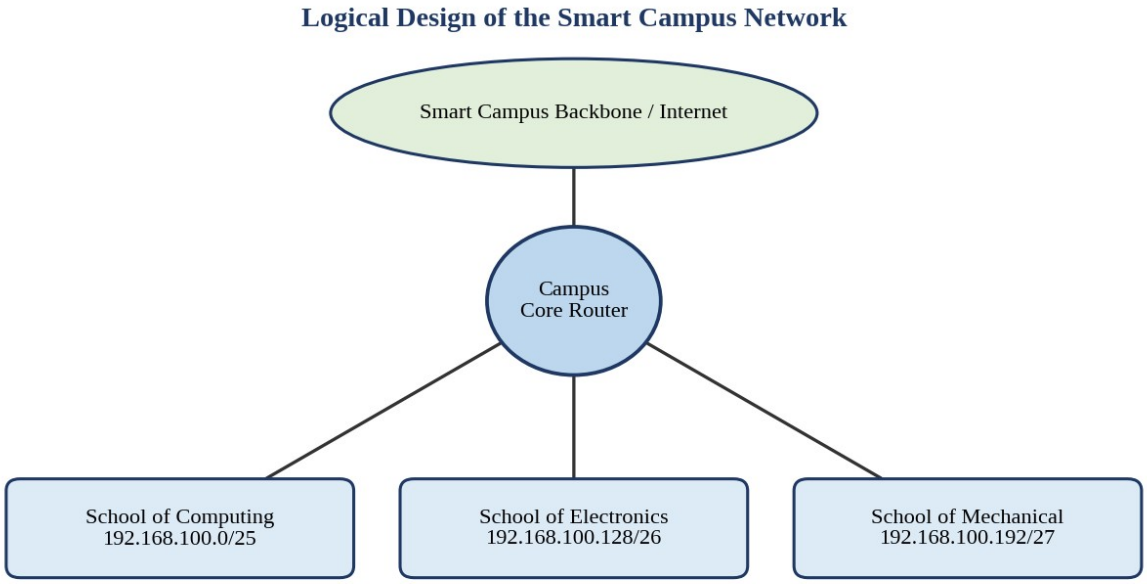


Figure 1: Logical design: one router interface (and one subnet) for each department

2. Identification of Key Issues

- The three subnets have **different sizes**, so VLSM must be used instead of equal-sized subnets.
- Each network address must be a multiple of its own block size (128, 64 and 32), otherwise the subnets would overlap or become invalid.
- The block must be used efficiently so that free space remains for future expansion.

3. Application of Concepts

$$\text{Block size} = 2^{(32 - \text{prefix})} \qquad \text{Usable hosts} = \text{block size} - 2$$

/25 → 7 host bits → block 128 → 126 usable hosts. **/26** → 6 host bits → block 64 → 62 usable hosts. **/27** → 5 host bits → block 32 → 30 usable hosts.

Computing. Start at 192.168.100.0. Broadcast = 0 + 128 − 1 = 127. **Electronics.** The next free address is .128, a multiple of 64. Broadcast = 128 + 64 − 1 = 191. **Mechanical.** The next free address is .192, a multiple of 32. Broadcast = 192 + 32 − 1 = 223.

Department	Network address	Subnet mask	Valid host range	Broadcast	Usable hosts
School of Computing	192.168.100.0/25	255.255.255.128	192.168.100.1 – 192.168.100.126	192.168.100.127	126

Department	Network address	Subnet mask	Valid host range	Broadcast	Usable hosts
School of Electronics	192.168.100.128/26	255.255.255.192	192.168.100.129 – 192.168.100.190	192.168.100.191	62
School of Mechanical Eng.	192.168.100.192/27	255.255.255.224	192.168.100.193 – 192.168.100.222	192.168.100.223	30

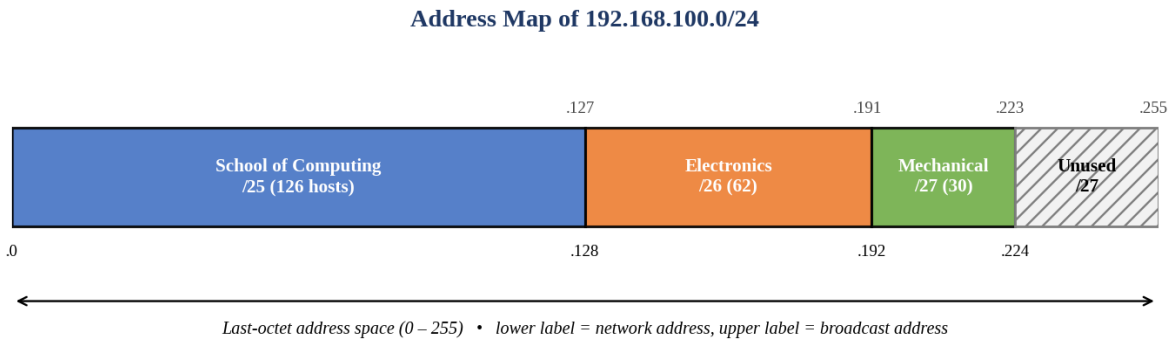


Figure 2: Address map showing the three departments and the free block

Unused range. The last broadcast address used is 192.168.100.223, so the remaining range is **192.168.100.224 – 192.168.100.255** (32 addresses, equal to the block 192.168.100.224/27 with 30 usable hosts).

4. Decision Making

The largest subnet (Computing) is placed first and the smallest last, so every boundary is aligned and the free space stays as one continuous /27. This free /27 is deliberately left unallocated; it can later be assigned to a new department, a server farm or a router-to-router link without changing any existing address.

Result and Interpretation: The three subnets use 128 + 64 + 32 = 224 addresses without overlap and provide 126, 62 and 30 usable hosts. The range 192.168.100.224 – .255 remains free, which is 12.5% of the block.

Answer 2: Subnet Allocation for the Software Company Using 10.10.0.0/16

1. Understanding of the Scenario

The company headquarters has been given the private Class A block **10.10.0.0/16**, which contains 65,536 addresses. Four departments need subnets of different sizes: **Software Development → /24**, **Quality Assurance → /25**, **Human Resources → /26** and **Executive Management → /27**. The addresses of each subnet and the free space left in the block must be determined.

2. Identification of Key Issues

- The /16 block has a large reserve, so the main task is to keep the departments in a compact, well-aligned group and leave the remaining space contiguous.
- The /24 subnet uses a full third octet, whereas the /25, /26 and /27 subnets divide the **fourth** octet, so the third octet changes from 0 to 1 after the first subnet.
- Subnets must be allocated largest-first to avoid overlap and fragmentation.

3. Application of Concepts

Software Development (/24). Host bits = 8, block size 256, usable hosts = $2^8 - 2 = 254$. The first block is 10.10.0.0 – 10.10.0.255, so the network address is 10.10.0.0 and the broadcast is 10.10.0.255.

Quality Assurance (/25). Host bits = 7, block size 128, usable hosts = $2^7 - 2 = 126$. The next free address is 10.10.1.0, so the broadcast is 10.10.1.(0 + 128 − 1) = 10.10.1.127.

Human Resources (/26). Host bits = 6, block size 64, usable hosts = $2^6 - 2 = 62$. The next free address is 10.10.1.128 and the broadcast is 10.10.1.191.

Executive Management (/27). Host bits = 5, block size 32, usable hosts = $2^5 - 2 = 30$. The next free address is 10.10.1.192 and the broadcast is 10.10.1.223.

Department	Network address	Subnet mask	Valid host range	Broadcast	Usable hosts
Software Development	10.10.0.0/24	255.255.255.0	10.10.0.1 – 10.10.0.254	10.10.0.255	254
Quality Assurance	10.10.1.0/25	255.255.255.128	10.10.1.1 – 10.10.1.126	10.10.1.127	126
Human Resources	10.10.1.128/26	255.255.255.192	10.10.1.129 – 10.10.1.190	10.10.1.191	62
Executive Management	10.10.1.192/27	255.255.255.224	10.10.1.193 – 10.10.1.222	10.10.1.223	30

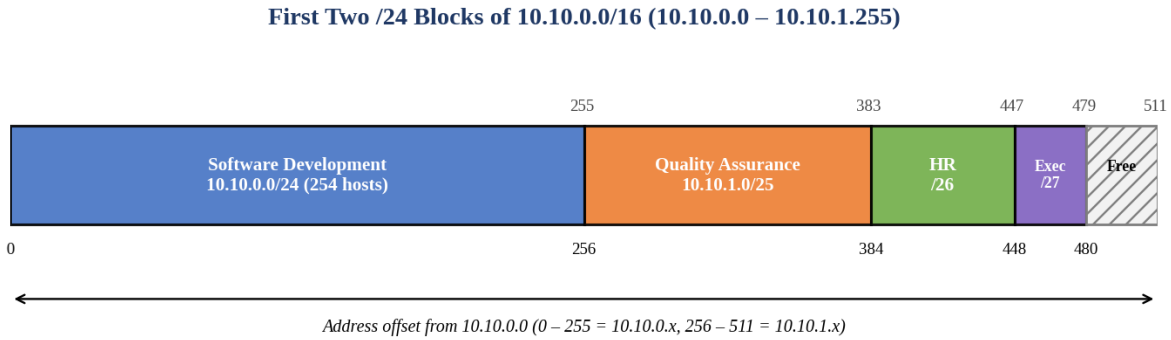


Figure 3: Allocation of the four subnets at the start of the 10.10.0.0/16 block

Unused range. The last address used is 10.10.1.223, so the unused space is **10.10.1.224 – 10.10.255.255**. Addresses used = 256 + 128 + 64 + 32 = 480, so unused addresses = 65,536 − 480 = **65,056**. This free space begins with the block 10.10.1.224/27 and continues with all of 10.10.2.0 – 10.10.255.255.

4. Decision Making

Placing the four subnets at the very start of the /16 block keeps routing tables simple, because one summary route, 10.10.0.0/23 (covering 10.10.0.0 – 10.10.1.255), can represent the whole headquarters. The rest of the /16 is saved for future sites, so the company can grow to hundreds of additional subnets without renumbering.

Result and Interpretation: Software Development (254 hosts), Quality Assurance (126), Human Resources (62) and Executive Management (30) are placed without overlap in 10.10.0.0 – 10.10.1.223. The free range 10.10.1.224 – 10.10.255.255 holds 65,056 addresses.

Answer 3: IPv6 Migration Plan for the Smart Hospital (2001:db8:abcd::/64)

1. Understanding of the Scenario

The hospital is moving thousands of medical devices and IoT sensors from IPv4 to IPv6. It has been assigned the network **2001:0db8:abcd:0000:0000:0000:0000:0000/64**. The engineer must write the address in compressed and expanded forms, identify the prefix and calculate how many addresses the subnet can support.

2. Identification of Key Issues

- IPv4 offers far too few addresses (a /24 gives 254) for thousands of sensors, which is the main reason for migration.
- IPv6 text notation is long, so correct use of the compression rules is needed to avoid errors in documentation and configuration.
- A /64 prefix is the standard size of an IPv6 LAN, because automatic address configuration (SLAAC) needs a 64-bit interface identifier.

3. Application of Concepts

(1) Compression. Remove the leading zeros in every group, then replace the longest run of all-zero groups with "::" (used once only).

```
Full form             : 2001:0db8:abcd:0000:0000:0000:0000:0000
Drop leading zeros    : 2001:db8:abcd:0:0:0:0:0
Replace 0:0:0:0:0 -> :: : 2001:db8:abcd::       (compressed form)
With prefix length    : 2001:db8:abcd::/64
```

(2) Expansion. The compressed form shows three explicit groups (2001, db8, abcd). Therefore "::" represents 8 − 3 = 5 groups of zeros, and every group is padded to four hexadecimal digits:

```
2001:db8:abcd::  ->  2001:0db8:abcd:0000:0000:0000:0000:0000
```

(3) Network prefix. With /64, the first 64 bits (four groups) identify the network: 2001:0db8:abcd:0000, written **2001:db8:abcd::/64** (also 2001:db8:abcd:0::/64). The last 64 bits (four groups, all zero here) are the interface identifier that individual devices will use.

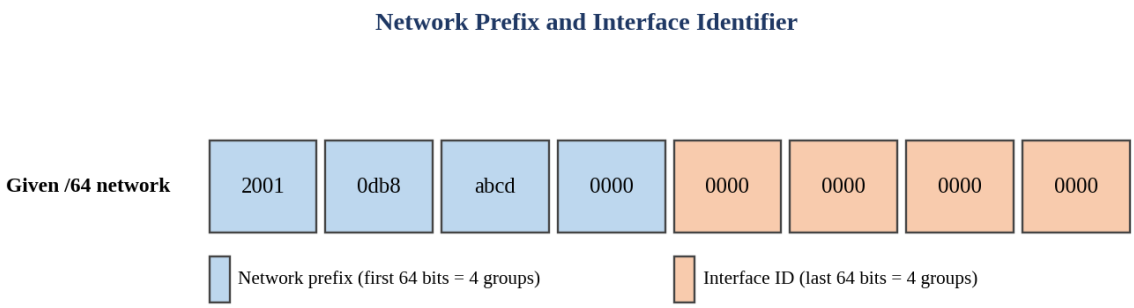


Figure 4: Division of the hospital network address at the /64 boundary

(4) and (5) Host capacity. Host bits = 128 − 64 = 64. The number of addresses is:

$$2^{64} = 18{,}446{,}744{,}073{,}709{,}551{,}616 \approx 1.8 \times 10^{19}$$

Items 4 and 5 of the task both ask for the host capacity of this network, so the same value applies: **18,446,744,073,709,551,616 addresses**. IPv6 has no broadcast address, so nothing is subtracted for it. One address (interface identifier all zeros) is reserved as the subnet-router anycast address, so $2^{64} - 1$ addresses can be assigned to devices.

4. Decision Making

A /64 is chosen for each device group (for example one for medical equipment and one for sensors) because it supports SLAAC and gives more than enough space. If the hospital receives a /48 site prefix, it can create $2^{16} = 65,536$ such /64 subnets, so every ward or system can have its own isolated subnet for security.

Result and Interpretation: The network compresses to 2001:db8:abcd::/64 and expands back to 2001:0db8:abcd:0000:0000:0000:0000:0000. The prefix is 2001:db8:abcd::/64 and the subnet can address 18,446,744,073,709,551,616 devices, which easily supports thousands of IoT devices.

Answer 4: Packet Forwarding Decision at the ISP Core Router

1. Understanding of the Scenario

An ISP core router connects three customer networks and holds three routes: **192.168.1.0/24**, **192.168.2.0/24** and **192.168.3.0/24**. A packet with the destination **192.168.2.45** arrives. The router must find the destination subnet, select the matching route, forward the packet to the correct next hop and use the default route if nothing matches. The scenario does not list interfaces or next-hop addresses, so the following are **assumed** for the explanation: customer A on Gi0/0 (next hop 10.0.1.2), customer B on Gi0/1 (10.0.2.2), customer C on Gi0/2 (10.0.3.2) and the ISP upstream on Gi0/3 (203.0.113.1).

2. Identification of Key Issues

- A router always chooses the route with the **longest prefix match**, so the packet must be compared with every entry.
- The three prefixes differ only in the third octet, so that octet decides the match.
- A default route is required for destinations that are not listed, otherwise the packet is dropped.

3. Application of Concepts

(1) Destination subnet. 192.168.2.45 lies in the /24 network 192.168.2.0/24 (network 192.168.2.0, broadcast 192.168.2.255, hosts .1 – .254). **(2) Matching entry.** Comparing the first 24 bits in binary:

```
Destination 192.168.2.45: 11000000.10101000.00000010.00101101
192.168.1.0/24          : 11000000.10101000.00000001 -> NO match
192.168.2.0/24          : 11000000.10101000.00000010 -> MATCH (24 bits)
192.168.3.0/24          : 11000000.10101000.00000011 -> NO match
```

Only the entry **192.168.2.0/24** matches, with a prefix length of 24. **(3) Forwarding.** Under the stated assumption, the router forwards the packet through **interface Gi0/1** to the next hop **10.0.2.2**, which leads to customer B’s network.

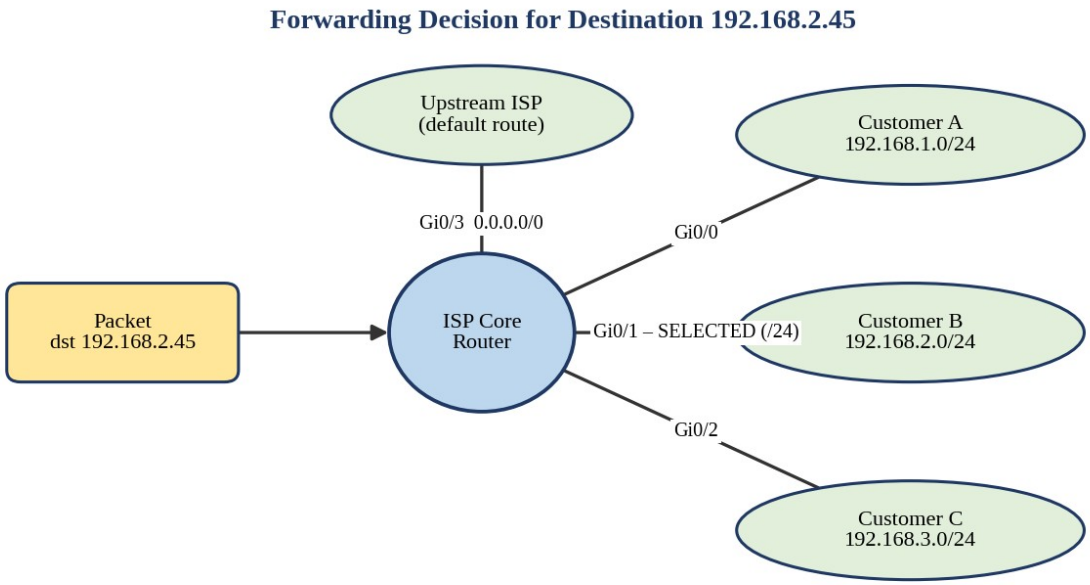


Figure 5: Forwarding decision of the core router for the packet addressed to 192.168.2.45

(4) Default route. If the destination matches none of the three entries (for example 8.8.8.8), the router uses the default route **0.0.0.0/0** (prefix length 0, the lowest priority), pointing to the upstream gateway 203.0.113.1 (assumed). If no default route exists, the packet is discarded and an ICMP "destination unreachable" message is returned.

Cisco Packet Tracer configuration used to reproduce the table

```
Router(config)# ip route 192.168.1.0 255.255.255.0 10.0.1.2
Router(config)# ip route 192.168.2.0 255.255.255.0 10.0.2.2
Router(config)# ip route 192.168.3.0 255.255.255.0 10.0.3.2
Router(config)# ip route 0.0.0.0 0.0.0.0 203.0.113.1  ! default route
Router# show ip route 192.168.2.45  ! verification
```

4. Decision Making

The longest-prefix-match rule gives a deterministic and loop-free decision: the more specific route always wins over the default route. Keeping a default route toward the upstream provider ensures that traffic to the Internet continues to work without listing every external network.

Result and Interpretation: The packet for 192.168.2.45 belongs to 192.168.2.0/24, matches only that routing entry, and is forwarded out of Gi0/1 to next hop 10.0.2.2 (assumed addressing). Any destination without a match is sent by the default route 0.0.0.0/0.

Answer 5: Addressing and Gateways for the Administration and Finance LANs

1. Understanding of the Scenario

A company has two departments joined by a Layer-3 router: **Administration LAN 192.168.10.0/26** (gateway 192.168.10.1) and **Finance LAN 192.168.10.64/26** (gateway 192.168.10.65). The administrator must determine the address limits of both LANs, suggest host addresses for three computers in each LAN and state which default gateway each LAN must use.

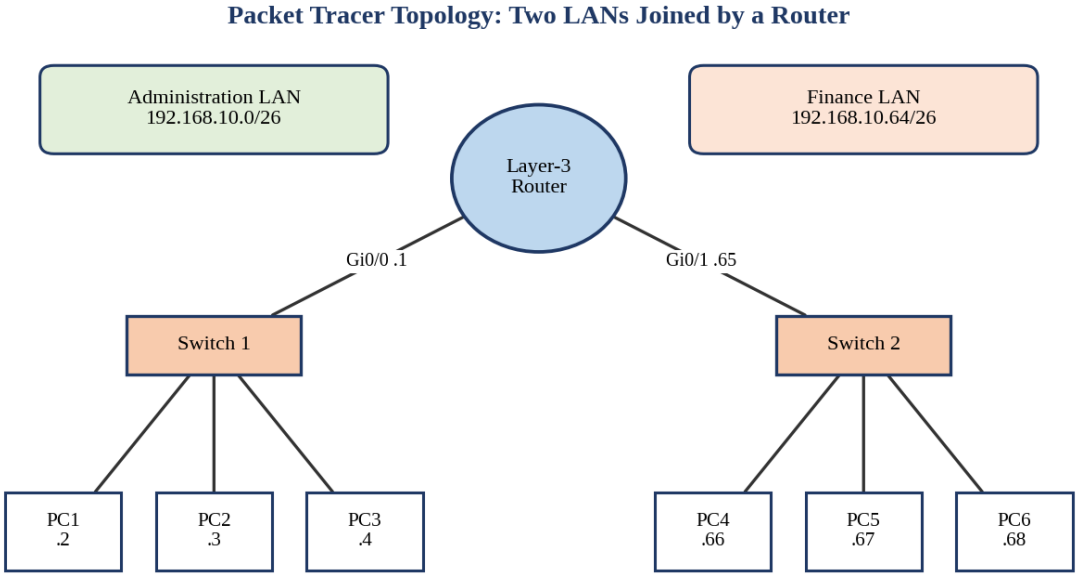


Figure 6: Topology of the Administration and Finance LANs (host addresses are shown by their last octet)

2. Identification of Key Issues

- Both LANs use a /26 mask (255.255.255.192, block size 64), so the networks begin at 192.168.10.0 and 192.168.10.64 and do not overlap.
- Hosts in different subnets can communicate only through the router, so each host must point to the router interface that lies in **its own** subnet.
- The gateway and the hosts of a LAN must share the same subnet, and no host may use the network or broadcast address.

3. Application of Concepts

Department	Network address	Subnet mask	Valid host range	Broadcast	Usable hosts
Administration LAN	192.168.10.0/26	255.255.255.192	192.168.10.1 – 192.168.10.62	192.168.10.63	62
Finance LAN	192.168.10.64/26	255.255.255.192	192.168.10.65 – 192.168.10.126	192.168.10.127	62

Calculation. Administration: network 192.168.10.0, broadcast 0 + 64 − 1 = 63, hosts .1 – .62. Finance: network 192.168.10.64, broadcast 64 + 64 − 1 = 127, hosts .65 – .126. Each

LAN offers $2^6 - 2 = 62$ usable addresses. The gateways 192.168.10.1 and 192.168.10.65 are the first usable addresses of their LANs, so both are valid.

Device	IP address	Subnet mask	Default gateway
Admin PC1	192.168.10.2	255.255.255.192	192.168.10.1
Admin PC2	192.168.10.3	255.255.255.192	192.168.10.1
Admin PC3	192.168.10.4	255.255.255.192	192.168.10.1
Finance PC4	192.168.10.66	255.255.255.192	192.168.10.65
Finance PC5	192.168.10.67	255.255.255.192	192.168.10.65
Finance PC6	192.168.10.68	255.255.255.192	192.168.10.65

Router configuration and verification in Cisco Packet Tracer

```
Router(config)# interface gigabitEthernet 0/0
Router(config-if)# ip address 192.168.10.1 255.255.255.192
Router(config-if)# no shutdown
Router(config)# interface gigabitEthernet 0/1
Router(config-if)# ip address 192.168.10.65 255.255.255.192
Router(config-if)# no shutdown
PC> ping 192.168.10.66  ! from Admin PC1 to Finance PC4
Reply from 192.168.10.66: bytes=32 time<1ms TTL=127  ! expected result
```

4. Decision Making

The first usable address of each LAN is reserved for the router, and hosts start from the next addresses (.2 – .4 and .66 – .68), which leaves a clear, consistent plan for later growth. Hosts in the Administration LAN must use **192.168.10.1** and hosts in the Finance LAN must use **192.168.10.65** as their default gateway. A ping across the LANs is routed, so its TTL falls from 128 to 127.

Result and Interpretation: Administration: 192.168.10.0 – 192.168.10.63 (hosts .1 – .62, gateway .1). Finance: 192.168.10.64 – 192.168.10.127 (hosts .65 – .126, gateway .65). The suggested computers can communicate within their own LAN directly and with the other LAN through the router.

		g of the scenario and context			
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned , logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand